

Scribble: Auto-generated 2D avatars with diverse and inclusive art direction

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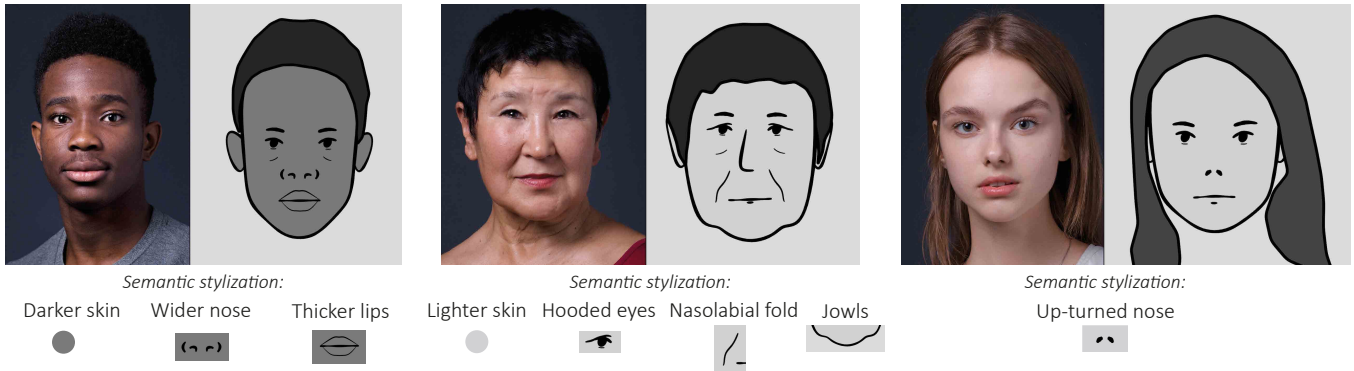


Figure 1: Example 2D stylized avatars automatically generated from a frontal selfie with diverse semantic stylization. Portrait photos from Infinite Realities.

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1 INTRODUCTION

Stylized avatars are a popular digital, graphical representation of unique users in teleconferencing, hybrid presence and gaming applications. As avatar systems become commonplace, there will be an increasing need to ensure that those systems support the wide range of appearance present in the human population. Existing human avatar systems typically use a single, generic base rig that morphs into varied human face and body shapes. It is costly to add extra anatomical shapes for under-represented human populations on top of a base rig originally designed for a majority demographic. This leaves many avatar systems struggling to catch up when it comes to fair representation of different demographics, especially for stylized avatars with limited visual language to depict diverse human attributes. We need new avatar systems, visual languages, and human understanding technologies that are co-designed with thoughtful consideration of diversity, right from the start.

Existing work in this domain has targeted photo-realistic avatars [Hu et al. 2017; Nagano et al. 2018] which fall into the uncanny valley, or 3D stylized avatars [Sang et al. 2022; Wang et al. 2023] with little consideration for diversity in their designs. We build a stylized face avatar rig with a visual style that accommodates diverse human features in its original design. We design a style mapping defining how diversity in human face shapes can be depicted stylistically, inspired by artistic references. For automatic avatar generation from a selfie, we develop new semantic detection and stylization techniques that specifically target the diverse features we set out to faithfully capture in our style mapping. We show 2D avatars generated automatically from a single frontal selfie, stylized explicitly based on our diversity-first art direction.

2 METHOD

Our system takes a single frontal face image as input and reconstructs the 3D face shape using the method of Wood et al. [2022]. Our rig generates a 2D avatar that stylizes the face shape and wrinkles, and then adds 2D hair enrolled from the input image. Our 2D avatar rig samples the identity mesh directly and generates 2D curves depicting general shape, and relative proportions of the head, ears, eyes, eyelids, eyebrows, nose, and mouth. For certain parts of the face we use differently styled curves for diverse identities. We show the template rig and identity variation before semantic stylization in Figure 2.

Facial Features. We choose to explicitly stylize certain facial features inspired by how various 2D illustrators depict diverse faces in appealing ways. We map variety in human eye, nose, chin, wrinkle, and lip shapes to unique stylized depictions of these traits in 2D as shown in Figure 3.

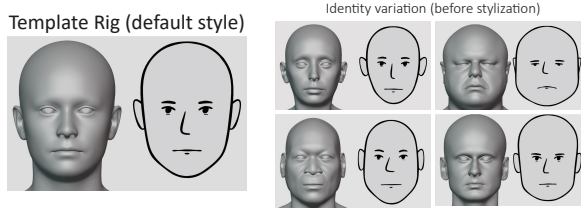


Figure 2: Template avatar rig with identity variation before stylization.

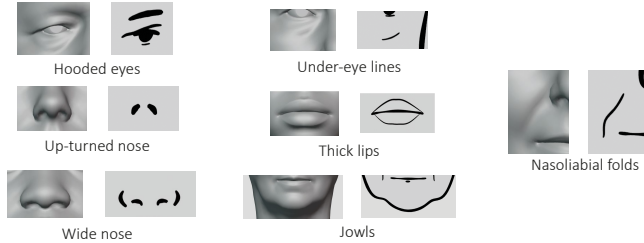


Figure 3: Our designed style mapping showing which diverse facial features we explicitly detect and how we stylize them in an appealing manner.

We use a part-based principle component analysis (PCA) decomposition of the face model to detect semantic facial features. To learn this semantic parameterization, we compute the PCA decomposition of eyes, nose, and chin regions of training samples of the face model of Wood et al. [2022]. We manually determine which learned PCA components activate in the presence of the relevant features in the identity mesh, if these pass certain thresholds which are determined manually then we render different curve styles.

For thick-lips, we simply measure the distance between the top and bottom of the lips in the solved identity mesh, and render the thick-lip curves if the distance exceeds 2.7 cm. While simple thresholding of distances and PCA coefficients can cause misclassifications, our rig still preserves likeness through relative proportions as all curves directly sample positions on the identity mesh.

Wrinkles. Curvature information can be utilized for wrinkle detection, as it allows us to identify areas where the mesh surface displays fine-scaled details often associated with wrinkles. Hence, for the nasolabial and under-eye regions, we compute the per-vertex curvature followed by the application of a hard threshold to the curvature values to identify wrinkles in those regions.

Additionally, we incorporate smoothing to leverage the contrast between wrinkle and non-wrinkle regions for detection. By comparing the vertex positions before and after applying the smoothing operation, we can identify the disparities caused by wrinkles. For each detected region, we check whether the difference between vertex positions before and after smoothing is higher than a certain threshold. This step further enhances the robustness of our wrinkle detection process. With these region-wise detection results we apply artist-designed wrinkle features to the appropriate regions of the avatar to preserve likeness while retaining stylistic consistency.

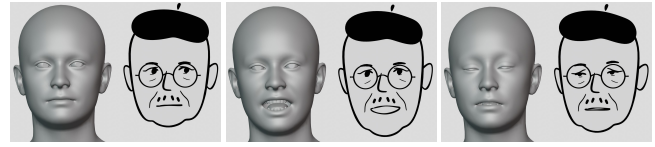


Figure 4: Frames of our avatars driven by facial animation supporting several diversity options in our rig including custom accessory assets.

Hair, Color and Accessories. Given the 2D domain of our avatars and assumption of frontal input images, we directly extract hair shape from the input image via segmentation. We train a DNN for semantic segmentation of the face as described by Wood et al. [2021], outputting classes for skin, head hair, facial hair, eyebrows, headwear and glasses.

Using these predicted semantic regions we use classical contour extraction [Suzuki and Abe 1985] to find the boundary of the applicable regions. These contours are post-processed to smooth the boundary to conform with the avatar style and normalized by head scale.

We extract the mean color of each region and map it to a gray-scale palette to approximate hair and skin color. We use the glasses segmentation output to select the closest style from a library of hand-authored glasses assets. Note that headwear (hats, headscarves) can be treated the same as hair to achieve visually compelling results for arbitrary appearances. Accessories can also be selected by the user from a library of artist-designed assets.

Animation. Our rig directly samples vertices from the face geometry of Wood et al. [2022] and so naturally supports head tracking by animating the identity mesh with the solved expression parameters. For 2D head motion we also apply the solved translation and pose parameters but scale down head rotation to avoid side profiles while preserving subtle head turns. In Figure 4 we show that our animation scheme supports all diverse stylization options in our rig, including eye-closing expressions for hooded eyes, and even custom face assets like hair, glasses, and hats.

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